



DATA. INSIGHTS. INTELLIGENCE.

Data Quality & Intelligence Report · Generated 11 July 2026, 13:51 · Coltrane Ltd

Executive Summary

Original Rows	974	Cleaned Rows	974
Rows Removed	0	Columns	20
Missing Values	2,504	Overall Risk	High
Top Issue	Missing (suffix)	Avg Issue Rate	11.82%

Bottom Line

This is a clean, well-structured patient/person registry of 974 unique individuals — usable right now for population demographics, geographic segmentation, and cohort analysis across the greater Boston area. The 12.9% overall missing rate looks alarming on paper but is concentrated in fields that are only expected to be populated for a subset of people (deathdate, suffix, maiden), not in core identity or demographic fields. Risk is primarily driven by structurally expected missing data, not data corruption.

Data Quality Summary

Missing values: 2,504 of 19,480 cells (12.9%).

Risk Level	Columns Flagged	Status
High	3	Action required
Medium	1	Review flagged
Low	19	Within tolerance

Average issue rate across columns: 11.82%.

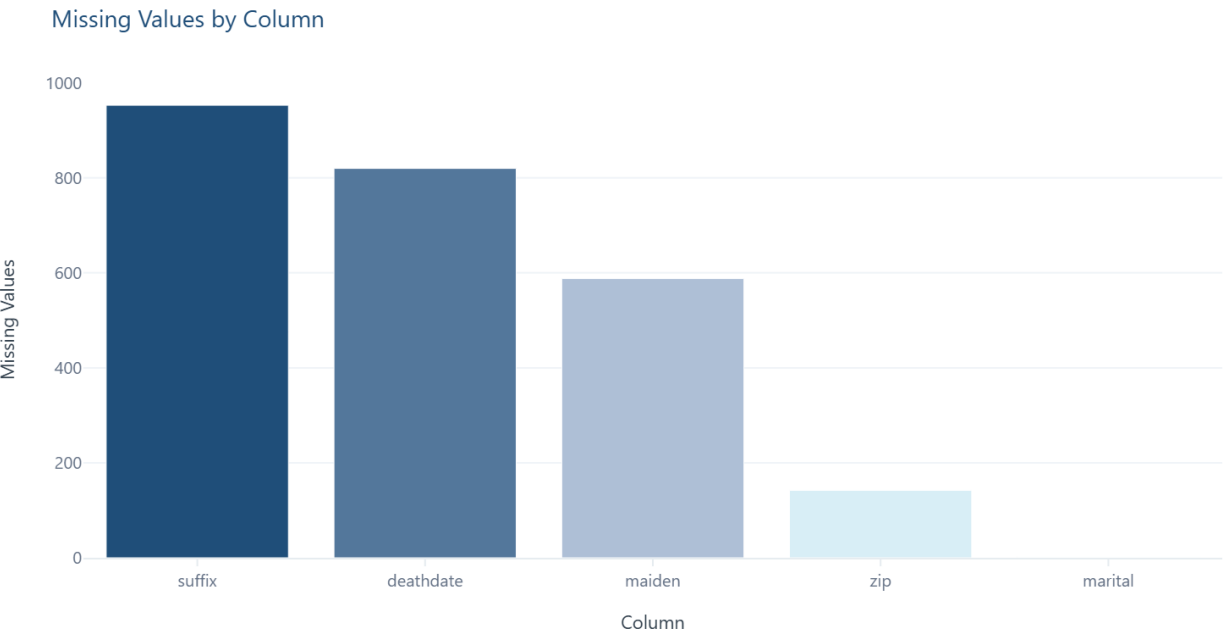
Risk is primarily driven by structurally expected missing data, not data corruption.

Visual Insights

Premium charts generated automatically from the cleaned dataset, mirroring the live dashboard and embedded in the Excel Executive Summary.

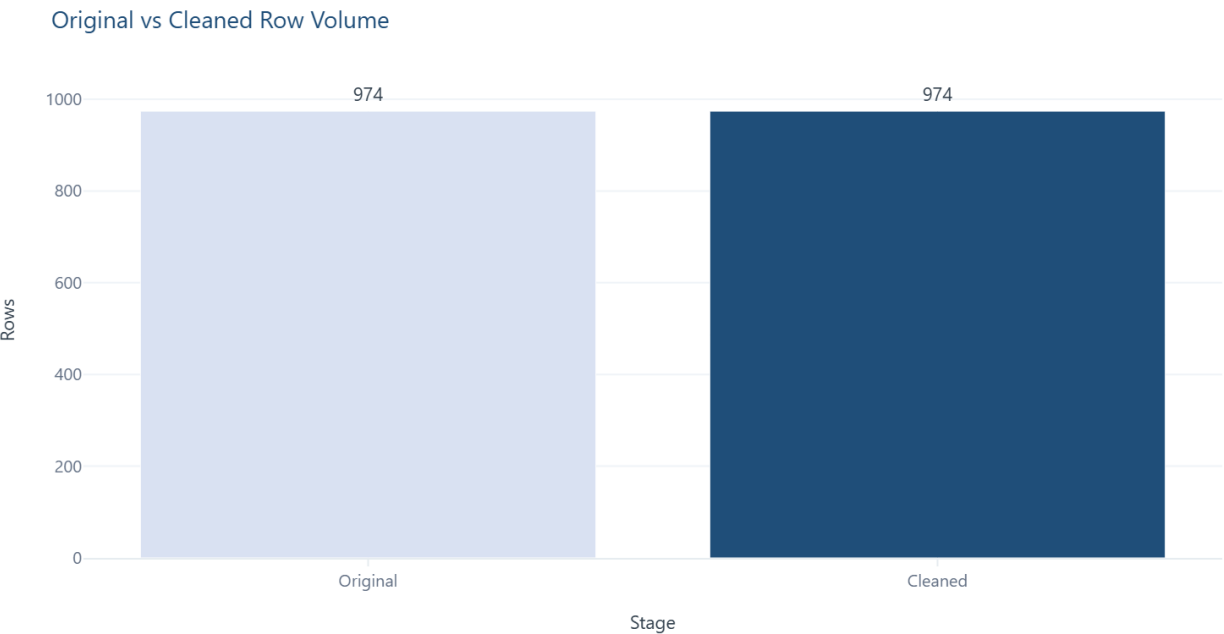
Missing Values by Column

Highlights which fields carry the most missing data after cleaning.



Original vs Cleaned Row Volume

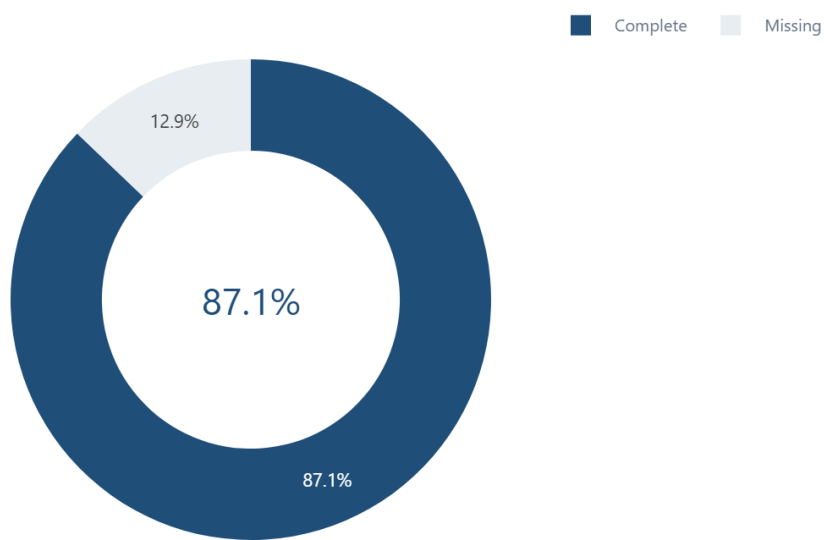
Shows how many records remain after duplicate removal and cleansing.



Overall Data Completeness

Proportion of populated cells across the cleaned dataset.

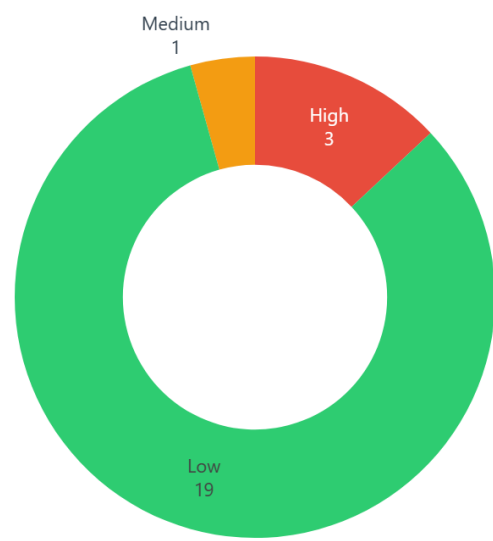
Overall Data Completeness



Column Risk Mix

Number of columns flagged at each data-quality risk level.

Column Risk Mix



Distribution — zip

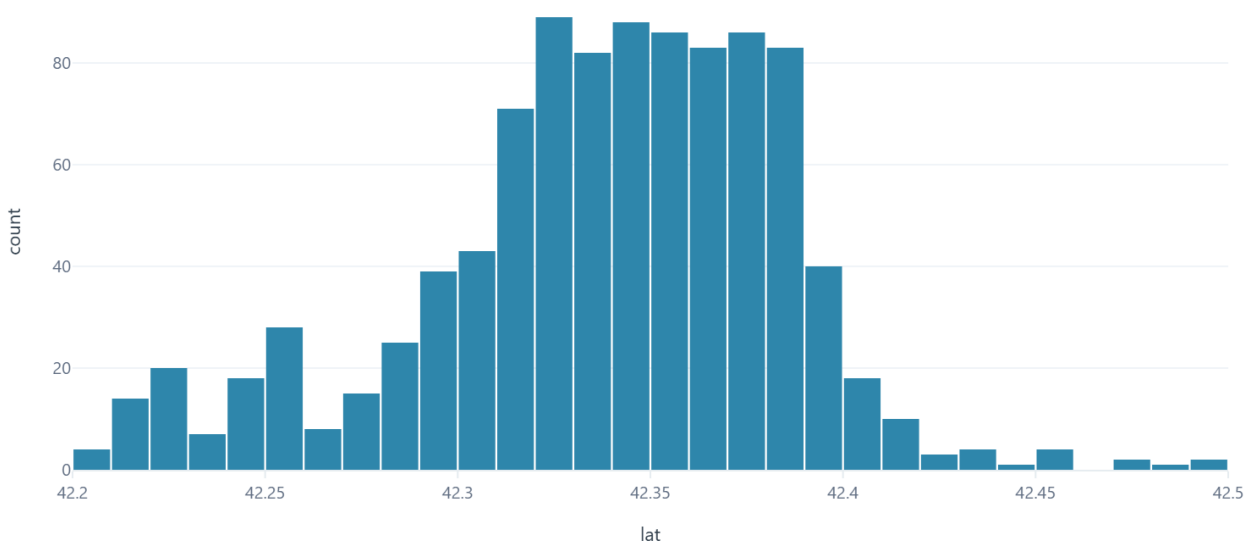
Frequency spread of values observed in 'zip'.

A histogram showing the distribution of zip codes. The x-axis is labeled 'zip' and ranges from 1800 to 2400 with major ticks every 100 units. The y-axis represents frequency, ranging from 0 to 500 with major ticks every 100 units. The distribution is highly right-skewed, with a very high frequency (nearly 500) for the bin centered around 2150. Other bins with non-zero frequency are located around 1850, 2000, 2050, 2100, 2200, 2400, and 2450.

zip bin (approx.)	Frequency (approx.)
1850	5
2000	25
2050	10
2100	490
2150	220
2200	50
2400	5
2450	35

Frequency spread of values observed in 'lat'.

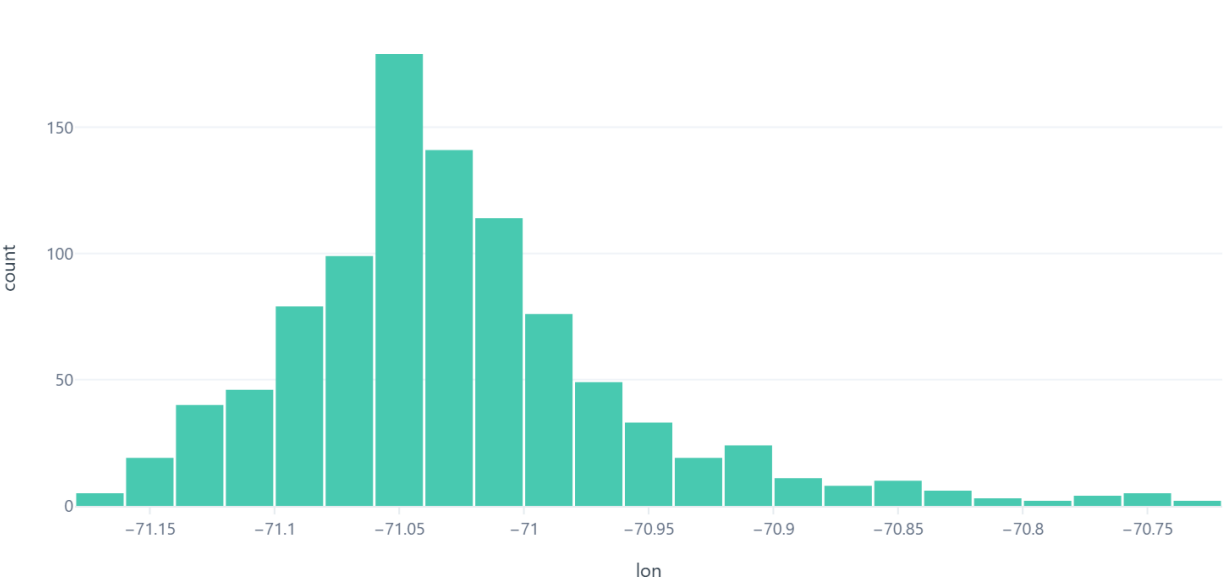
Distribution — lat



Distribution — lon

Frequency spread of values observed in 'lon'.

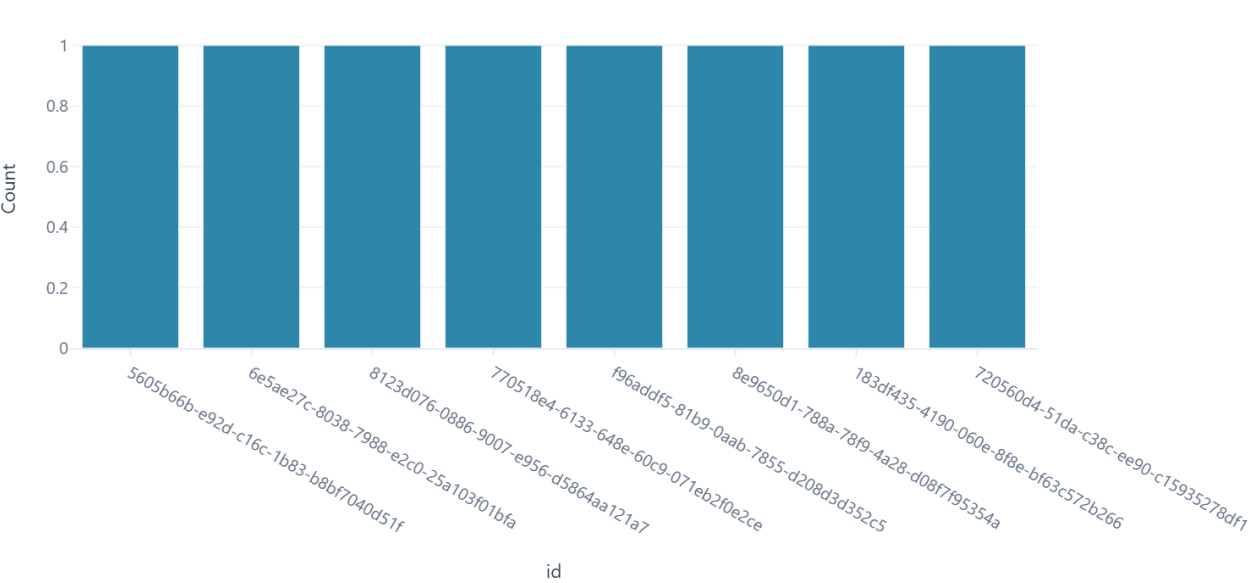
Distribution — lon



Top Categories — id

Most frequently occurring values in 'id'.

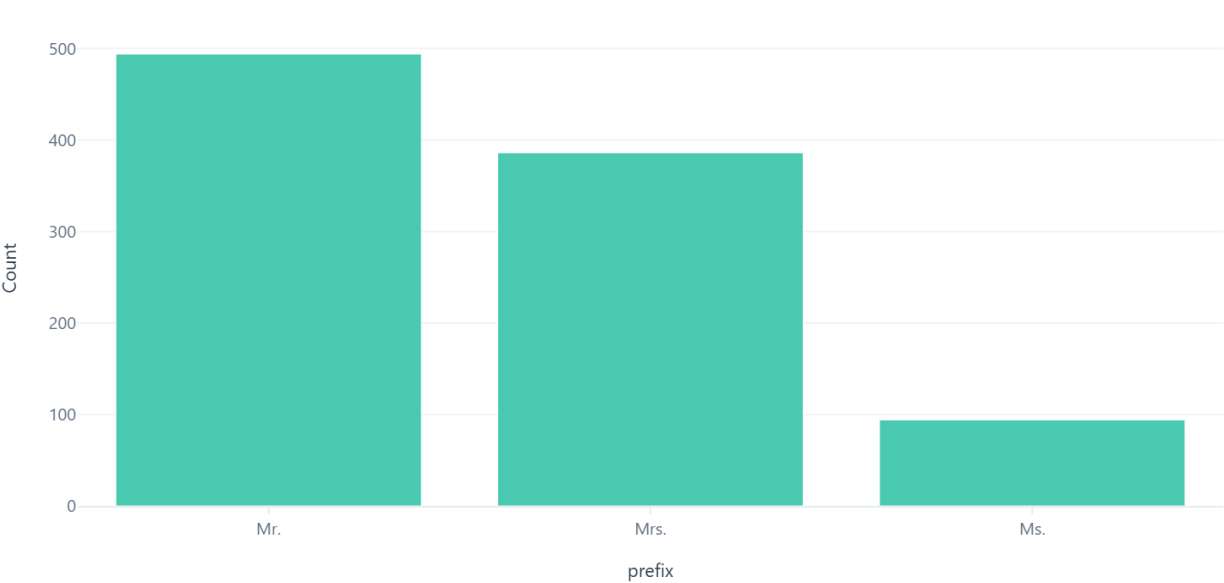
Top Categories — id



Top Categories — prefix

Most frequently occurring values in 'prefix'.

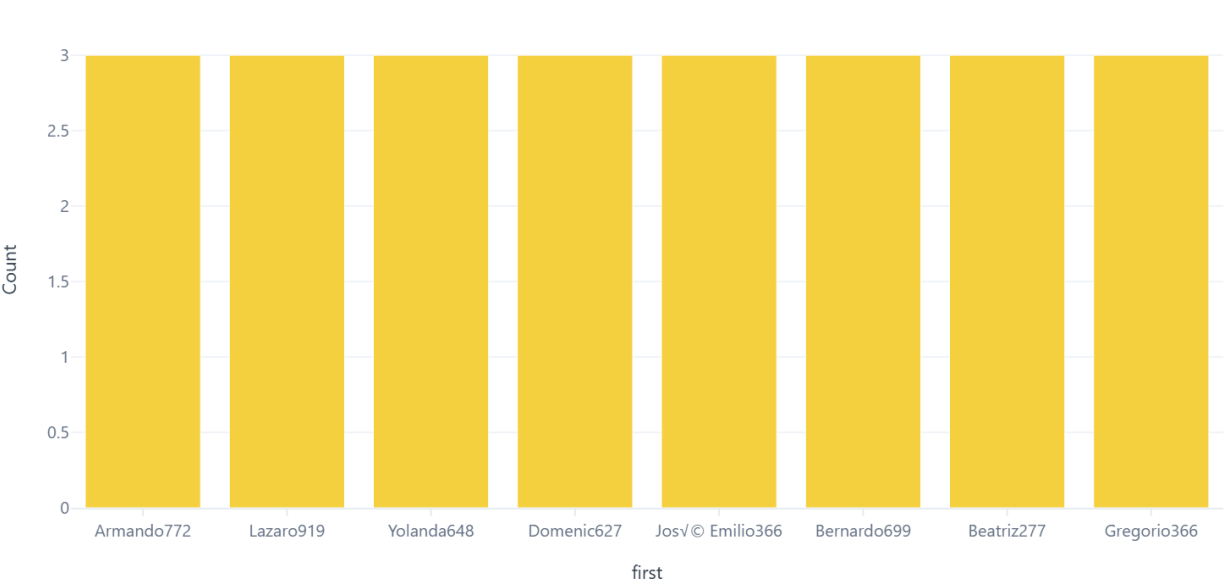
Top Categories — prefix



Top Categories — first

Most frequently occurring values in 'first'.

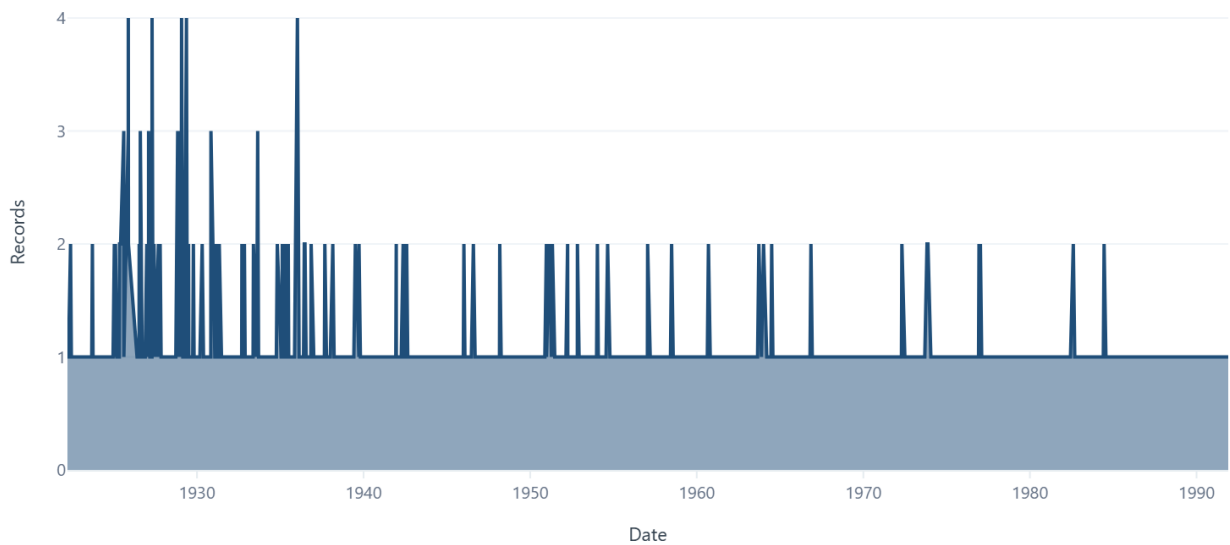
Top Categories — first



Records Over Time — birthdate

Volume of records by date — useful for spotting gaps, spikes or seasonality.

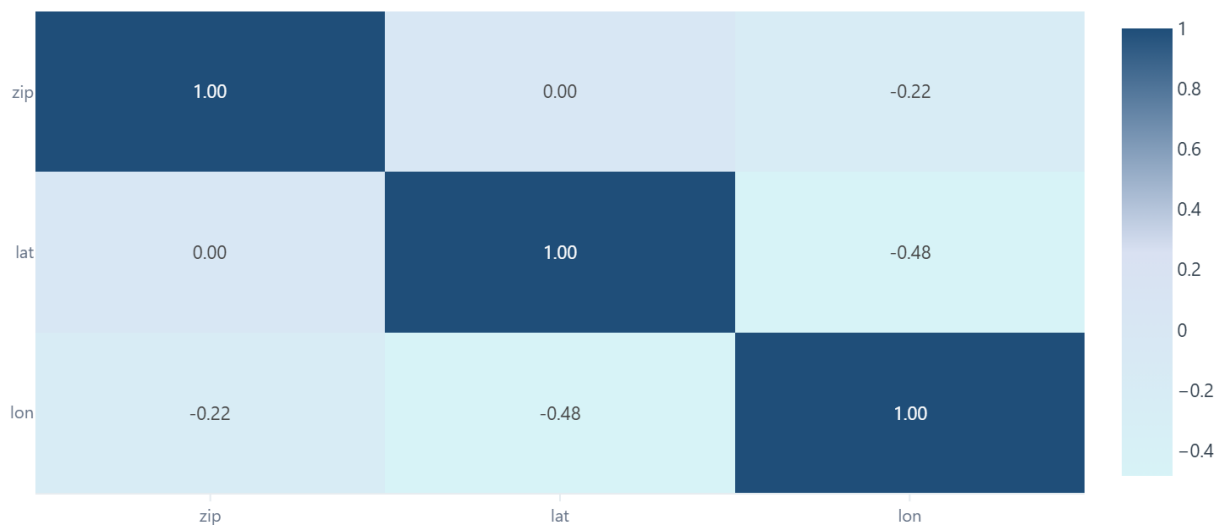
Records Over Time — birthdate



Correlation Between Numeric Fields

Pairwise relationships (Pearson correlation) between numeric columns.

Correlation Between Numeric Fields



Key Data Insights

Business-focused patterns drawn from the cleaned dataset — what is happening, and why it matters.

- **Married majority dominates the population.** marital is 'M' for 784 of 974 (80.5%) versus 'S' for 189. This is a settled, adult-skewed population — relevant for household-level products, insurance, or care planning where marital status drives eligibility and risk.
- **Demographics are heavily white/non-Hispanic.** race is 'white' for 680 (69.8%), 'black' 163 (16.7%), 'asian' 91 (9.3%); ethnicity is 'nonhispanic' for 783 (80.4%). Any model trained here will underrepresent

minority cohorts — a fairness and generalisability concern for downstream decisioning.

- **Gender is near-perfectly balanced.** gender splits 494 M / 480 F (50.7% / 49.3%). This is a rare clean signal and makes gender a reliable stratification variable without reweighting.
- **The population is elderly and partly deceased.** birthdate top values cluster in the late 1920s (1929, 1925), and only 148 of 974 (15.2%) carry a deathdate. This is a living-with-mortality-events registry — ideal for survival analysis and longitudinal outcomes, not for youth-oriented use cases.
- **Tight geographic footprint = single metro area.** lat ranges only 42.20–42.50 and lon –71.17 to –70.73, with zip from 01801 to 02472 — this is the Boston/eastern Massachusetts region. The dataset is geographically dense, which strengthens local modelling but prohibits national generalisation.
- **Names show synthetic/generated signatures.** Values like first = 'Armando772' and last = 'Heaney114' carry numeric suffixes — this is machine-generated (Synthea-style) data, not real PII. Commercially useful for safe development and testing, but not a real customer/patient base.
- **Prefix concentration reflects the marital and gender mix.** prefix is only 'Mr.' (494), 'Mrs.' (386), 'Ms.' (94) — consistent with 494 males and 480 females of whom most women are married. Prefix is redundant with gender+marital and adds little independent signal.

AI Advisory

Claude-powered interpretation of the cleaned data — patterns, anomalies, quality risks, and concrete next steps.

A. Data Quality Diagnosis

- **suffix (97.8% missing) and deathdate (84.2% missing) are structural, not broken.** Suffix (PhD/JD/MD) applies to a rare minority; deathdate only exists for the deceased 15.2%. Do not impute either — absence is meaningful information.
- **maiden (60.4% missing) is conditionally structural.** It only applies to individuals who changed a surname (predominantly married women). Presence correlates with marital='M' and gender='F'; treat missingness as a category, not a defect.
- **marital (0.1% missing) and zip (14.6% missing) are the only fields with genuinely repairable gaps.** The single missing marital is trivial; zip gaps can be recovered from lat/lon via reverse geocoding.
- **zip must never be treated as a numeric metric.** The summary reports zip mean=2152.42 and std=75.46 — this is meaningless. Zip is a categorical geographic code; a "mean zip" has no interpretation. Cast to string and preserve leading zeros (values like 01801 will otherwise corrupt to 1801).
- **id is a UUID key, not a feature.** All 974 values are unique — this is the primary key for joins and must be excluded from any statistical or ML feature set.
- **lat/lon are legitimate numerics** and are the only true continuous variables. Their near-zero variance (std 0.05 / 0.07) confirms a single-metro concentration rather than a data error.
- **Date fields are stored as strings.** birthdate and deathdate are categorical text (e.g. '1929-01-28') and must be parsed to datetime before any age, tenure, or survival calculation.

B. Statistical & Pattern Analysis

- **No meaningful continuous distributions to assess for shape.** The only true numerics (lat, lon) are spatially clustered near-constants; there is no revenue, cost, or usage variable to test for skew or heavy tails.

- **zip distribution is categorical, not statistical.** Its range (01801–02472) reflects the ZIP code numbering of eastern Massachusetts, not a quantitative spread — treat as ~50–80 discrete area codes.
- **No numeric outliers of concern.** Latitude/longitude bounds are tight and geographically valid; there are no extreme coordinate values suggesting bad rows.
- **Time coverage skews to early-20th-century birth cohorts.** Deathdates span 2012–2019, indicating recent mortality events on an aging population — a compressed, recent outcome window suitable for near-term survival modelling.
- **Name-value density is high and near-unique.** first (842 unique) and last (498 unique) approach row count; low collision confirms individual-level granularity with no aggregation.
- **Class balance varies sharply by field.** Gender is balanced (near 50/50), but race, ethnicity, and marital status are all heavily imbalanced — any classifier using these will need stratified sampling or class weighting.

C. Business Interpretation

- **This is a synthetic patient/person master record** — a clinical or demographic registry for the Boston metro, most consistent with a healthcare or life/health insurance domain.
- **Core commercial signal is demographic + geographic segmentation:** age (from birthdate), mortality status, race/ethnicity, marital status, and ZIP-level location together enable population health and actuarial cohorting.
- **The deceased subpopulation (148 records) is the highest-value analytical asset** — it enables outcome/survival modelling that the living majority cannot yet support.
- **Being synthetic is a feature for development.** This dataset is safe for building and validating pipelines, dashboards, and models without PII exposure — but insights must not be presented as real-population truths.
- **Geographic concentration limits commercial reach** but sharpens local relevance: ideal for a Massachusetts-focused product, unsuitable for national benchmarking.

Modelling & Analytics Readiness

This dataset is **decision- and cohort-analytics ready today**, and **near-model-ready** with modest transformation. Required steps and cautions:

- **Transformations required:** parse birthdate/deathdate to datetime and derive age and (where present) age_at_death; cast zip to zero-padded string; encode race, ethnicity, marital, gender as categoricals; drop id from features; add explicit is_deceased, has_maiden, has_suffix indicator flags instead of imputing.
- **Leakage risk — critical:** deathdate is a direct outcome. If you model mortality or survival, deathdate (and any age-at-death derivation) must be the target/label, never an input feature. Including it as a predictor guarantees perfect but useless leakage.
- **Secondary leakage:** suffix and maiden presence correlate with time-of-record and life events; using their raw presence to predict downstream outcomes can smuggle in future information — flag and audit.
- **Imbalance handling:** race/ethnicity/marital imbalance requires stratified splits and class weighting to avoid a model that simply predicts the majority class.
- **Recommended approaches:** (1) **Survival analysis / time-to-event (Cox or Kaplan–Meier)** using birthdate→deathdate for the deceased cohort; (2) **unsupervised clustering (k-modes / mixed-type)** on demographics + ZIP for population segmentation; (3) **geographic aggregation** at ZIP level for

population-health dashboards. Standard regression on this data has no continuous target and is not recommended.

Top Priority Actions

- Fix zip typing — cast to zero-padded string and stop computing statistics on it — because a "mean zip" of 2152 is corrupt and leading zeros are already being lost (Effort: Low).
- Parse birthdate and deathdate to datetime and derive age/age_at_death — because every cohort, survival, and demographic insight depends on it (Effort: Low).
- Convert structural missingness in suffix, deathdate, maiden into explicit indicator flags rather than imputing — because absence carries real meaning and imputation would fabricate signal (Effort: Low).
- Isolate deathdate as a protected label and quarantine it from feature sets — because it is a direct outcome and the single largest leakage risk in this dataset (Effort: Medium).
- Reverse-geocode the 14.6% missing zip values from lat/lon — because location is a core segmentation axis and the coordinates make recovery cheap (Effort: Medium).
- Apply stratified sampling / class weighting for race, ethnicity, marital — because 70% white and 80% non-Hispanic will otherwise bias any classifier (Effort: Medium).
- Label the dataset explicitly as synthetic in all downstream reporting — because insights must not be presented to the CEO as real-population fact (Effort: Low).

Bottom Line Summary

Decision-ready now, model-ready after light transformation — the data is structurally sound with no corruption; its High risk score is entirely a function of expected conditional-missingness in suffix, deathdate, and maiden, and once dates are parsed, zip is typed correctly, and deathdate is fenced off from features, this becomes a reliable base for survival and demographic segmentation modelling on the Boston metro population.

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